

BitCrew

Planned

Software Design Document

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1.0 INTRODUCTION

1.1 Purpose

The purpose of this SDD is to provide an overview and to describe the architecture/system-design of AI Schedule Planner, a project in development by Bit Crew. The intended audience is the Bit Crew development team and Professor Edmund Dantes.

1.2 Scope

The purpose of Planned is to provide a seamless user experience in managing daily tasks and events with calendar integration. The main difference it has with traditional schedule planners is its AI assistant feature, which will play a central role in providing the intended seamless experience. The AI Schedule Planner application is designed for iOS and Android devices so that users can manage their schedule on the go. It will integrate features from these software environments, notably notification reminders.

1.3 Overview

This SDD covers the structural design of Planned, notably including a general architectural design, class structure, and an interface design philosophy. It provides class diagrams divided into the main areas of domain, backend, database layer, and AI. In addition to these, this SDD provides a database schema for the Planned application.

1.4 Definitions and Acronyms

AI – artificial intelligence

FCM – firebase cloud messaging

FAQ – Frequently Asked Questions

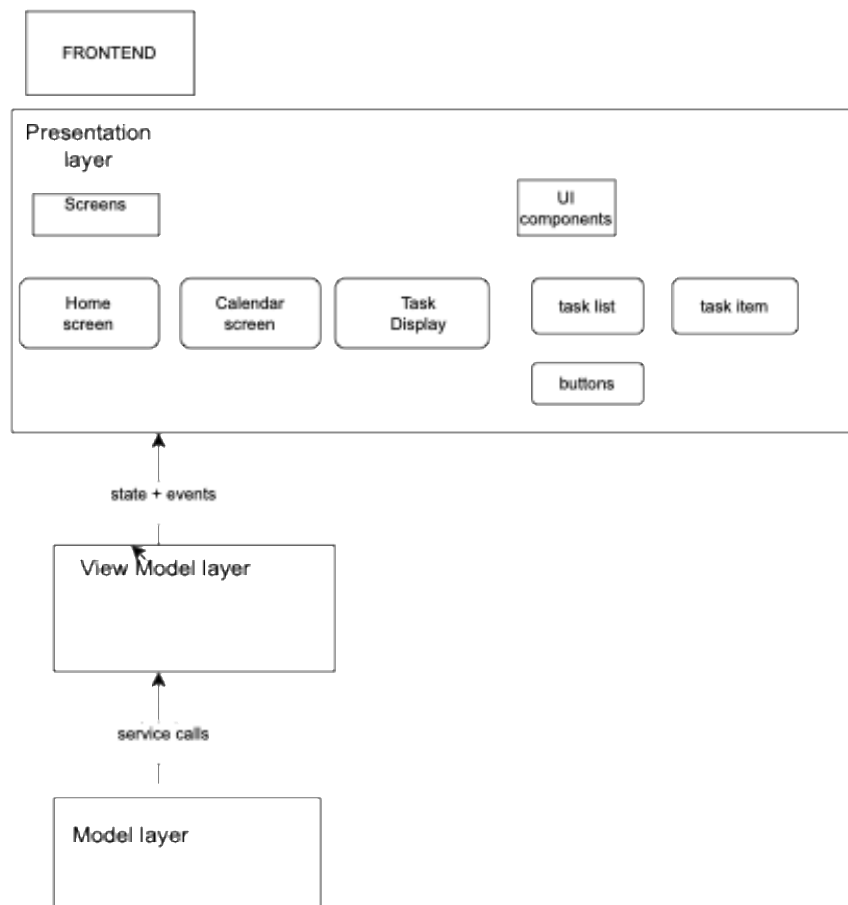
2.0 SYSTEM OVERVIEW

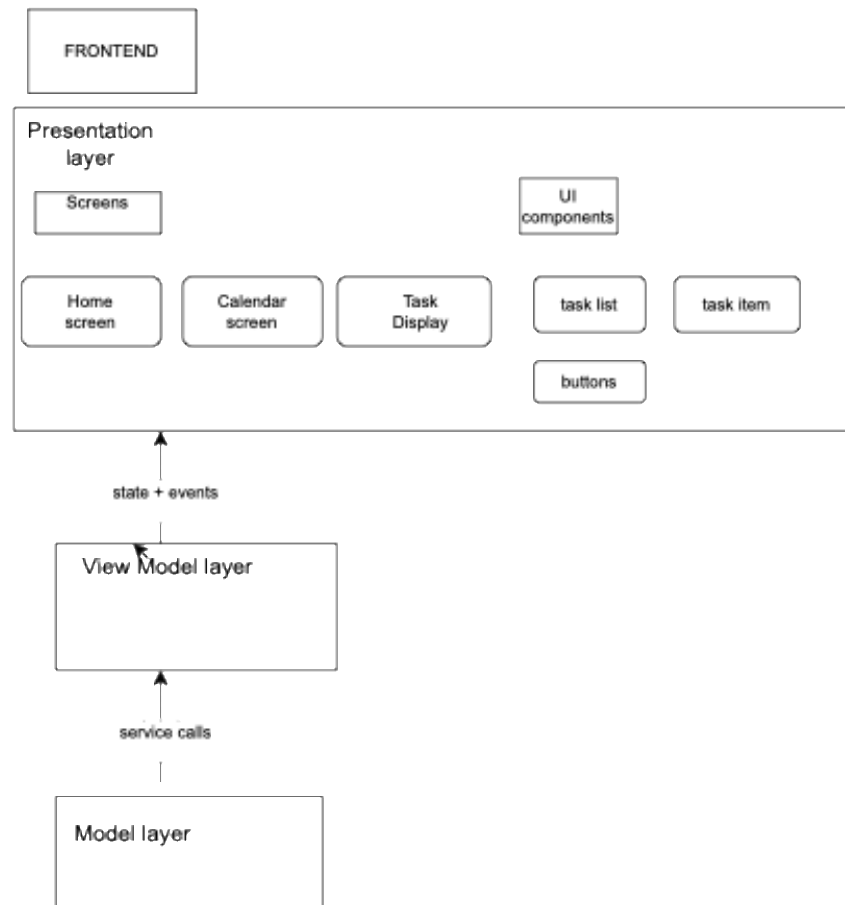
Planned is a mobile app for iOS (iPhone users) and Android users. This app allows students and busy people to manage tasks throughout the day, and based on user input, have AI create tasks as desired.

The idea behind this app is that time management can be very challenging for college students, so this alleviates the stress of having to remember all tasks by sending push notifications to remind when the task needs to be completed or when an event is planned.

3.0 SYSTEM ARCHITECTURE

3.1 Architectural Design





3.2 Interface Design

The model layer presents data and business logic. The view layer displays the data in the user's perspective as the UI. The ViewModel layer serves as a bridge to ensure a clear separation of concerns.

3.3 Decomposition Description

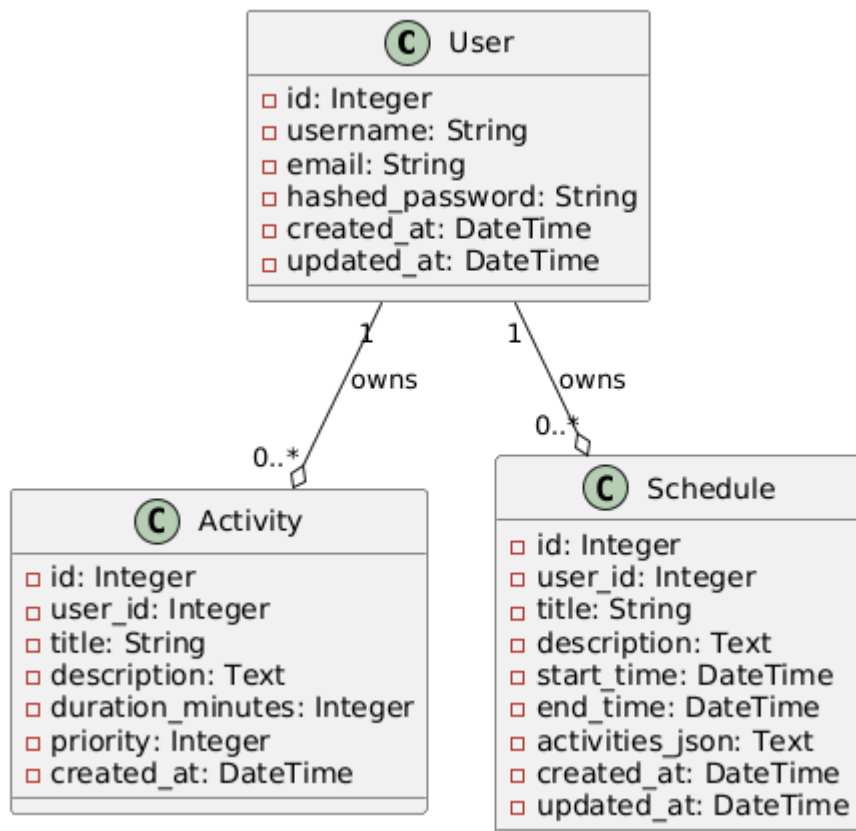
Our decomposition of the architectural design is discussed in this section. The presentation layer includes the screens- home screen, task display, and calendar and UI components- task list, task item, buttons. The view model is responsible for handling the UI's state and events. It deals with user actions, data transformation, and async calls. The model layer is responsible for its services: API communication with the FastAPI backend, handling all HTTP requests through the frontend service layer.

3.4 Design Rationale

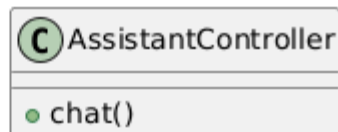
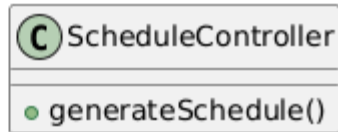
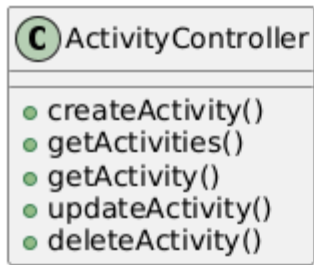
Our rationale for choosing layered architecture is discussed in this section. We chose the layered architecture for modularity, so each components can be separated and have little interference with each other. Each layer has its own responsibility so as to not have too much overlap with other layers. The architecture pattern is Model- view-view model for clear separation of concerns, testability, and maintainability. The view layer handles the UI where the user sees and interacts with the app, capturing input, displaying data, and managing the UI state. The viewModel layer is bridge between the View and Model. It manages view layer's data, handles view logic, and converts model data a user-friendly logic for the user to easily navigate. The model layer is responsible for managing the data, communication with other databases, local storage, and Fast API.. Data is validated to ensure consistency and validity.

4.0 COMPONENT DESIGN/DETAILED DESIGN

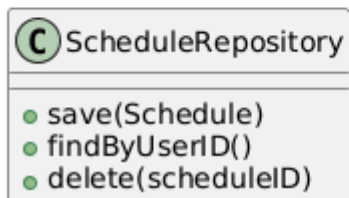
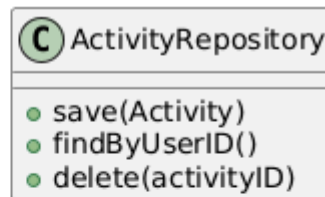
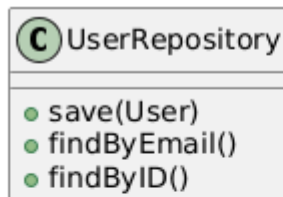
Domain

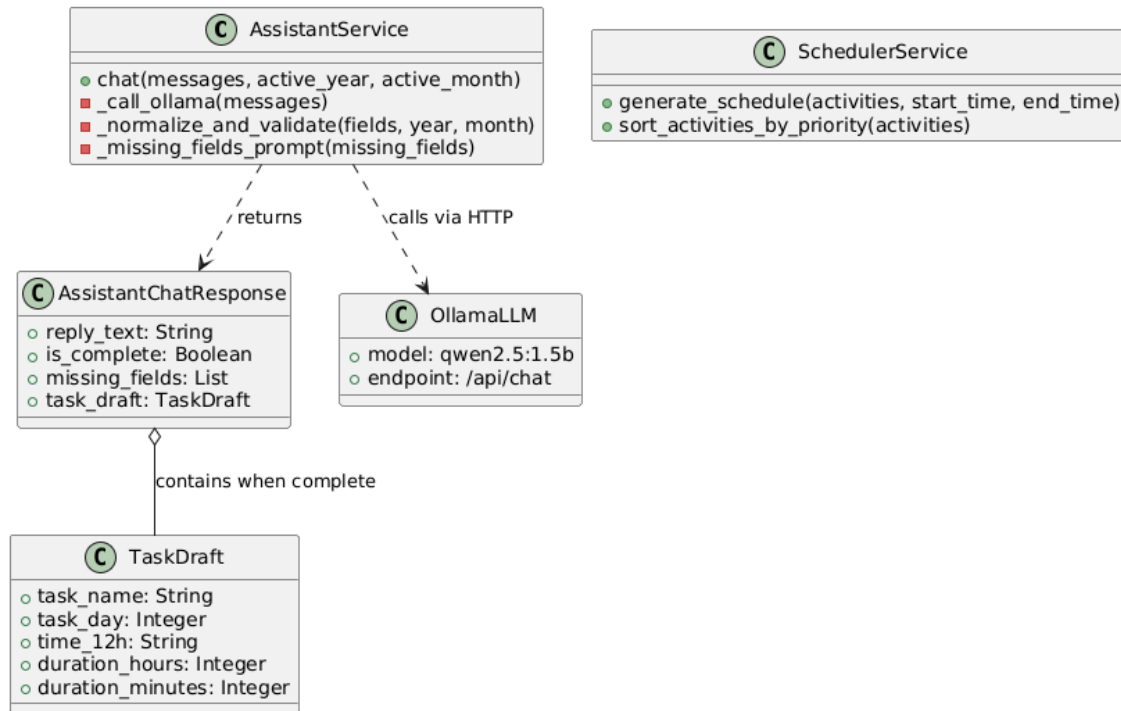


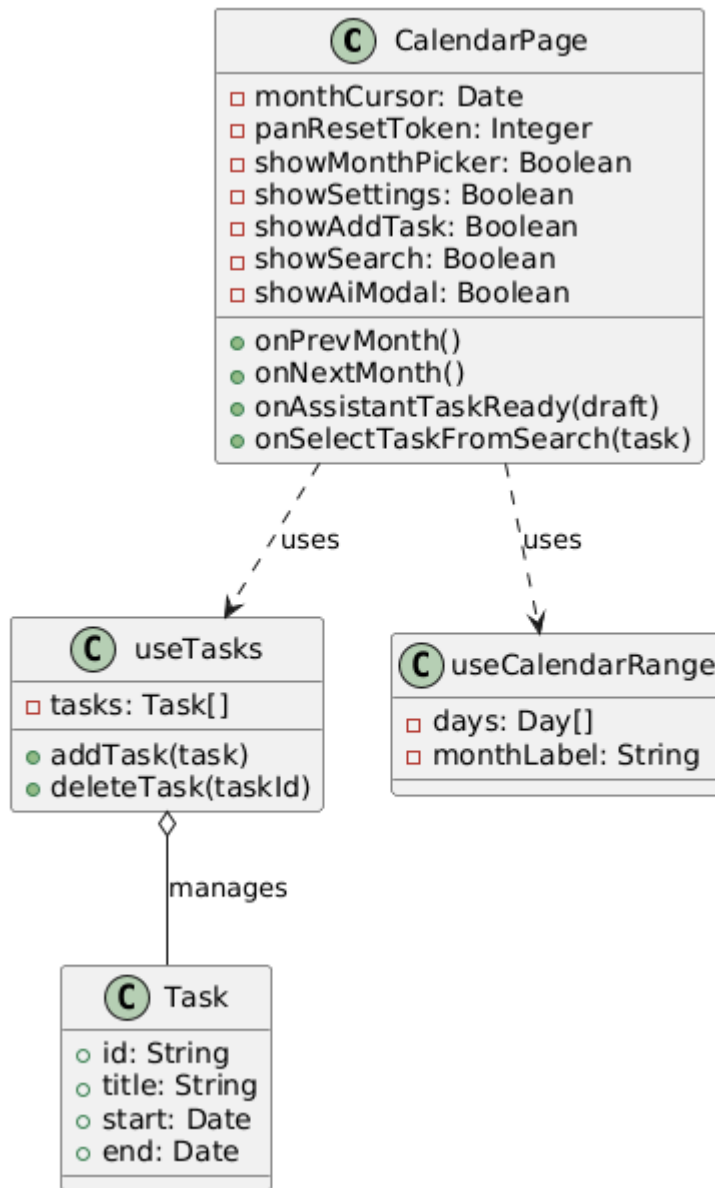
Backend



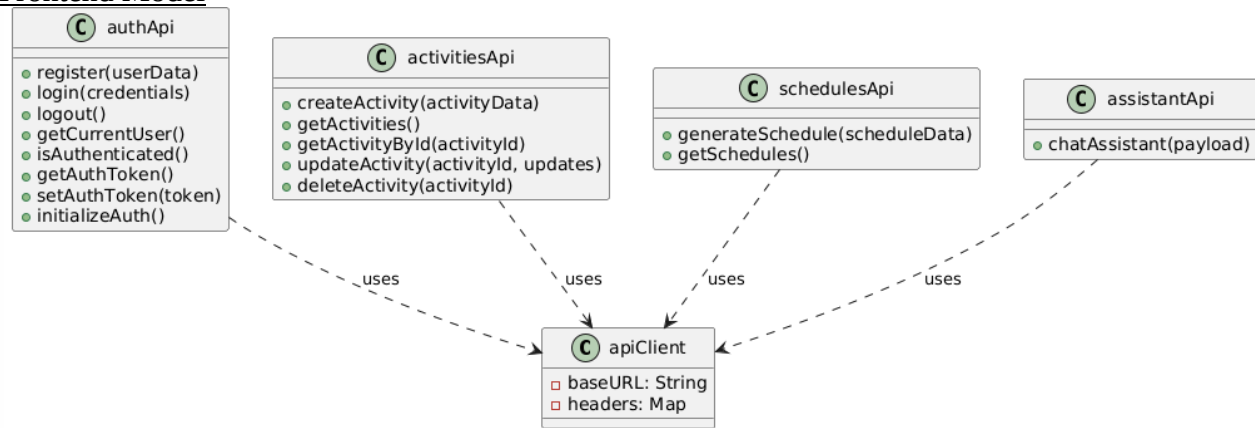
Database Layer



AI

Frontend ViewModel

Frontend Model

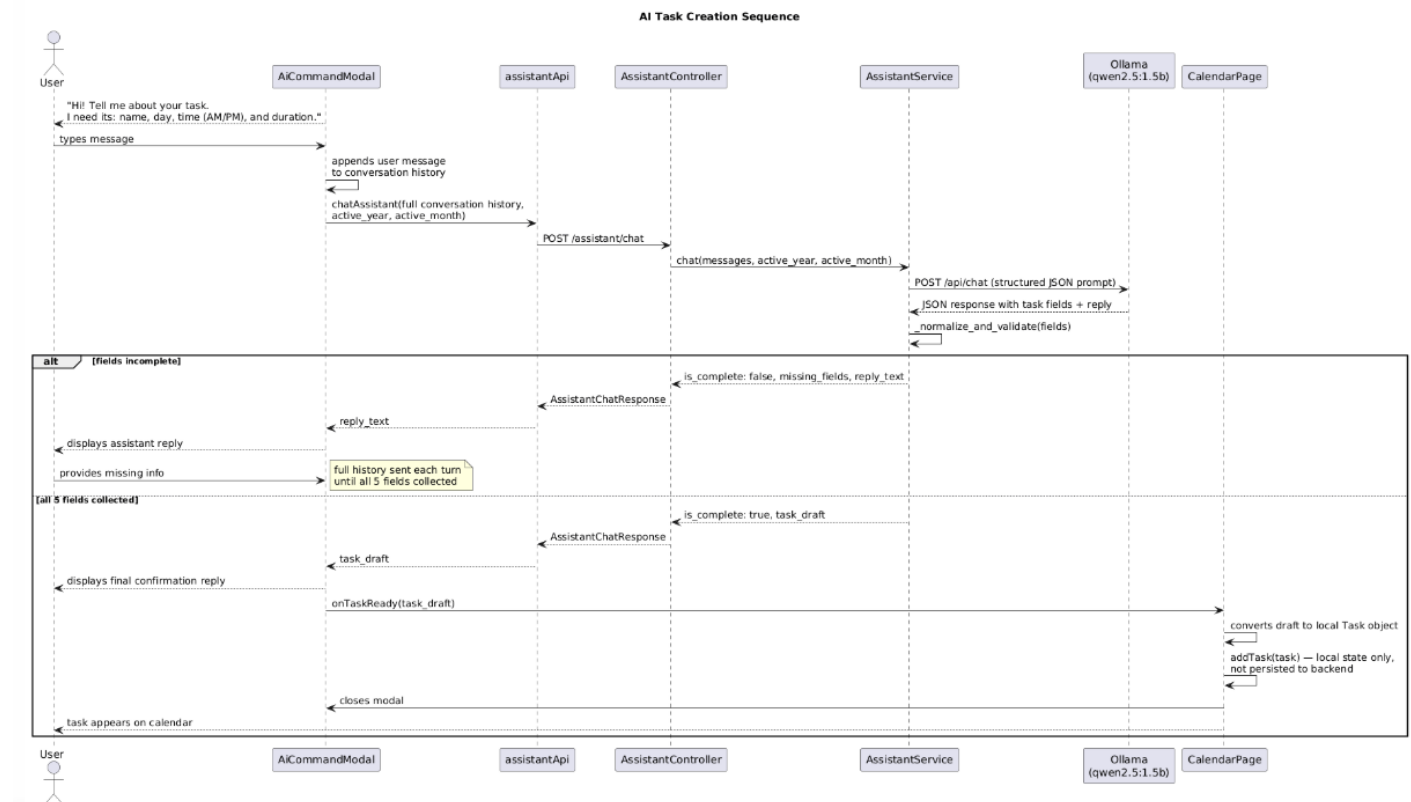


4.2 - Sequence Diagram

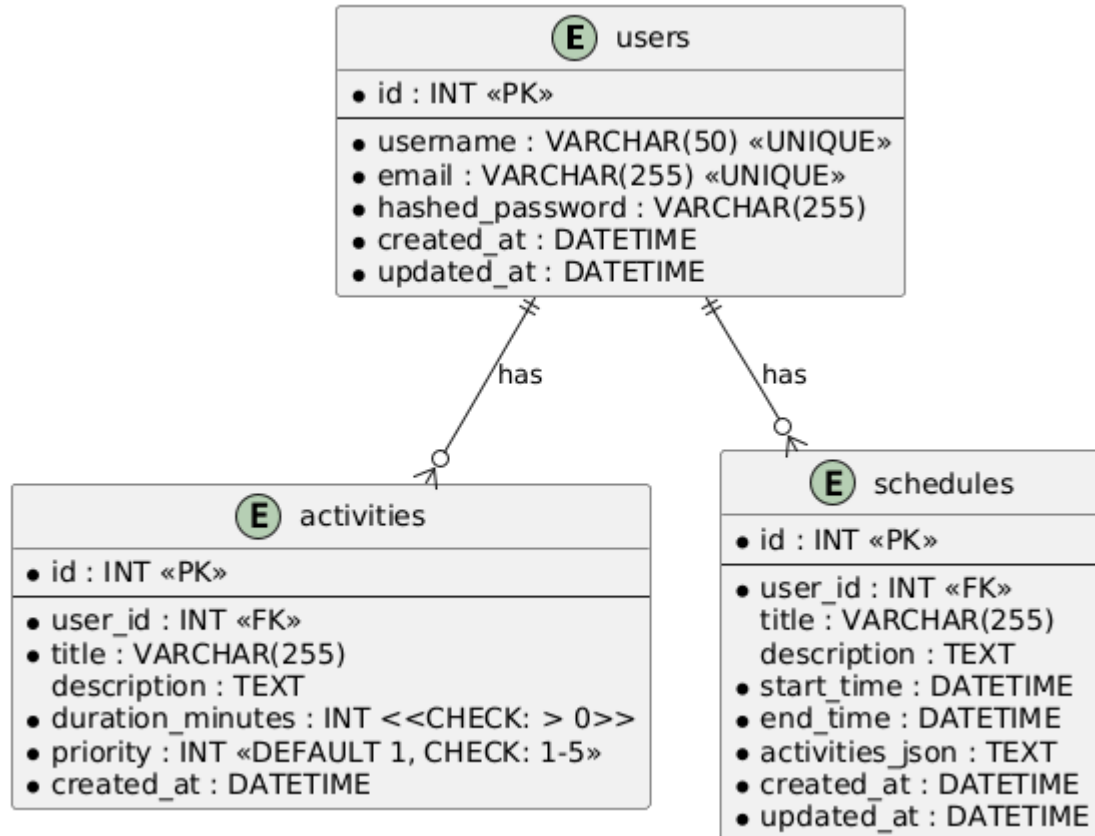
The below diagram displays how the system behaves over time when a user interacts with the AI assistant to create a task through guided conversation.

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https://www.plantuml.com/plantuml/png/fLJDRjiy4BpdAVRpoSQdE6Y1N7meOkXI5qrjnCxDWB0Mrp4RYbJ8bLFrr8Ve4_P9kg9-33XDukX94dPcbfpPrKrWz5mLvWA0DHk2-G8s61uXzOlidOKrFLLaSxeG2ERiFDm7yl9H2bFdkaJBC9hhr1K5MdNh59eHO83xnJa8Gz3IqFAyr14oNxt0p7jCwYnxPmpv5fg-YboJF-gSMjZw7BOq1WlSsl7JCzdtq-lPk-drRXB1oy_du1GDMOL-XG-jNhgiUyjLuFBoGxmup63qKVy76p862WBSkOgXTfK75j-cMxi0IwH0SvY1nO8IK5Wdm5hGu_djrUfsae0u1QhoqUBf17gquasJXRJWkgGWyY78GU81XXAM9LaLe19yZzXQTf0xUoGVeYGSn1hdwu4gJWitFo0FjettbJ5_P2LRAqdh8sKreK-W-oYSvSD494KiQgOYkLgkDt0r97Lj6WWQKIikwRZxg0XWRULrqAJN4x2fOwIC0xigvmhJme-hPTVeFIkABaX2VooRnPBdaBfR21urdo8iS1Uan7J_fTQQUgXK-Harla2ZVvEcSII7ULL8TEuvKpYu1lk9RJDFr4XRd81Q9LwDtJ8-keCzcY2f5zeCSS-P2q_QK-GCnTZNa2y6Y8wpR_S-gwxI0HsldTpS2vpFgHAXz9W7M28PhWwmDcqYPL7hIZqHmLjzWu3MiS4NZySU0PnOBePWK0YIPYBmPMtMrjPrWPG4DUzLncplZcJ4YqIBrwlksZ14z9Z35J7IV_PjiWynVRzjfYt8wzzaMx7IV_-grALq9nTYEQTuIg7f_K9nrG81smMpKE40703epBfKb31xVxAcQy9A5IJJqgJ-3NZvyTAJ0oIM_JRAjr32NvnW9IZV0EyqUoQf0ReiwD2qtGJg4PgeC1dJ_n7uC-Y3patHzHm8sWF6xaMHNcDm00



4.3 - Database Schema



5.0 COMPONENT DESIGN

5.1 Component Overview

The AI Schedule Planner app consists of the following major software components:

1. Authentication Component
2. User Profile & Preferences Component
3. Task Management Component
4. Calendar Component
5. Reminder & Notification Component
6. AI Scheduling Component
7. Help & Support Component

5.2 Component Descriptions

5.2.1 Authentication Component

Upon opening the app, the user is presented with the authentication screen. This component allows users to securely access their personal accounts. The options available on this screen include:

1. **Login:** Allows the user to enter their email and password to sign in. If credentials are invalid, an error message is displayed.
2. **Sign Up:** Takes the user to the account creation screen where they enter name, email, and password.
3. **Password Input:** Accepts secure password entry and hides characters as the user types.
4. **Authentication Handling:** Validates login data, processes credential checks, and establishes a secure session.

5.2.2 User Profile & Preferences Component

After authentication, the user's personalized profile data and scheduling preferences shape the way the app recommends tasks and schedules. This component includes:

1. **Profile Information:** Displays stored user data such as name and email.
2. **Preferences Input:** Allows the user to set profile information and notification settings.
3. **Data Storage:** Saves all preference updates and syncs them with the scheduling engine.

5.2.3 Task Management Component

The task management screen allows users to create, edit, and manage their activities. When the user opens this panel, the following options are displayed:

1. **+ New Task:** Opens the task creation form where users can input task name, due date, time, duration, and description.
2. **Task List:** Displays all upcoming tasks and allows users to tap on any task to view or modify it.
3. **Checkmark Button:** Marks a task as completed.
4. **Delete / Edit Options:** Accessible within each task's detail view.
5. **Task Metadata Display:** Shows due date, duration, and priority for each task.

5.2.4 Calendar Component

The calendar component provides the central scheduling interface. When users navigate to the calendar screen, the following elements are available:

1. **Monthly Grid View:** Displays all days of the month along with scheduled events or tasks.
2. **Daily Detail View:** When a day is selected, tasks and events for that date appear in a separate panel.
3. **Add Event Button:** Allows users to create new calendar events or link tasks to specific days.
4. **Navigation Arrows:** Move between months.

5.2.5 Reminder & Notification Component

This component manages alerts and reminders for deadlines, upcoming tasks, and schedule changes. When notifications are triggered, the following behavior occurs:

1. **Push Notifications:** Alerts the user of upcoming deadlines, conflict warnings, or AI-generated recommendations.
2. **Reminder Scheduling:** Automatically sets reminders for tasks based on due date and priority.
3. **Calendar Alerts:** Displays visual indicators (exclamation marks, highlight bars) inside the calendar grid.
4. **System Integration:** Utilizes expo-notifications + FCM for Android push notifications.

5.2.6 AI Scheduling Component

The AI assistant chats with the user and takes input to create the desired tasks. When the user interacts with this component, the following occurs:

1. **AI Response Window:** Displays the assistant's conversational replies as it collects task information from the user.
2. **User Message Input:** Allows the user to type natural language input describing their task. The assistant asks follow-up questions until all five required fields are provided.
3. **Task Field Collection:** The assistant collects task name, day, start time, duration hours, and duration minutes across one or more conversational turns.
4. **Task Draft Confirmation:** Once all five fields are collected, the system generates a task draft and places the task on the calendar without requiring further user action.

5.2.7 Help & Support Component

The Help & Support section provides user assistance and troubleshooting tools. When accessed, this panel displays:

1. **FAQ Panel:** Shows common help topics such as how to use the calendar, how to create tasks, and how the AI suggestions work.
2. **Contact Form:** Allows users to enter their name, email, and message to send support requests.
3. **Submit Button:** Sends the support message to the app's backend for processing.
4. **Tutorial Links:** Offers step-by-step guidance for app features.
5. **Support Feedback:** Notifies users when their message has been successfully submitted.

5.3 Screen Images

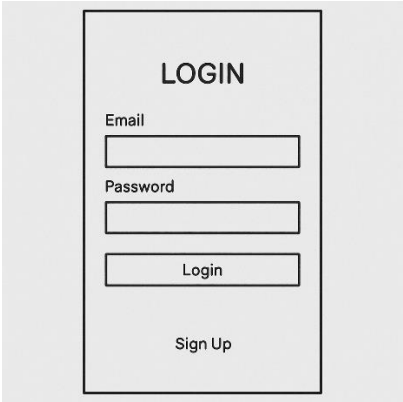


Fig.1 Login Screen

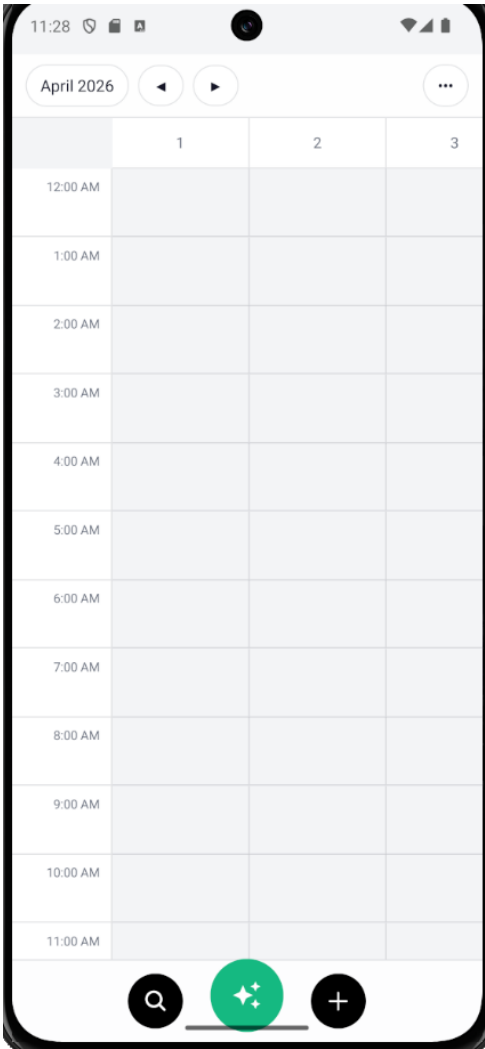


Fig.2 Calender Home Screen

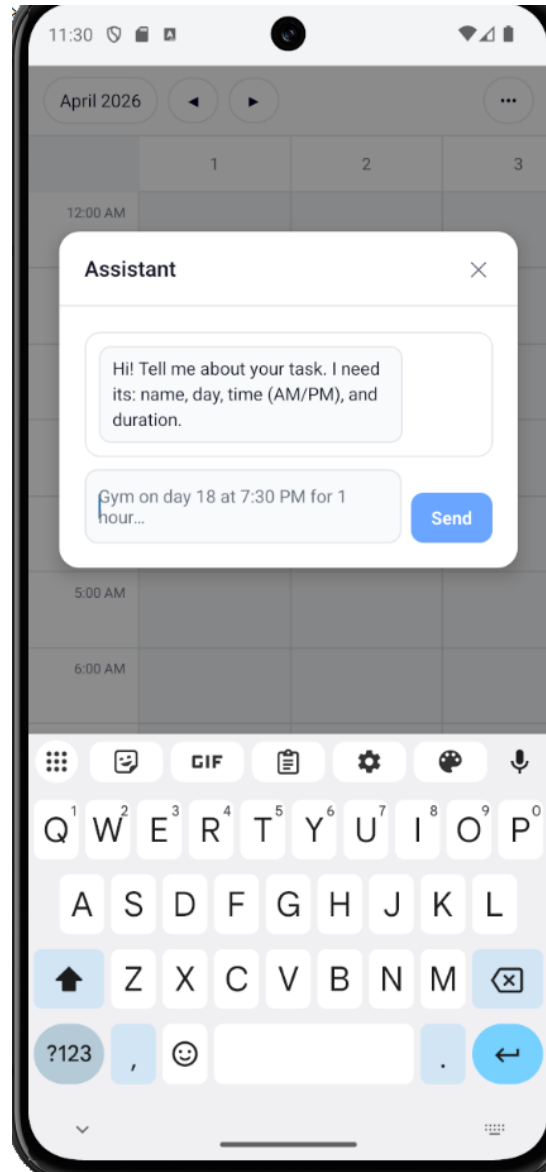


Fig.3 AI Assistant Screen

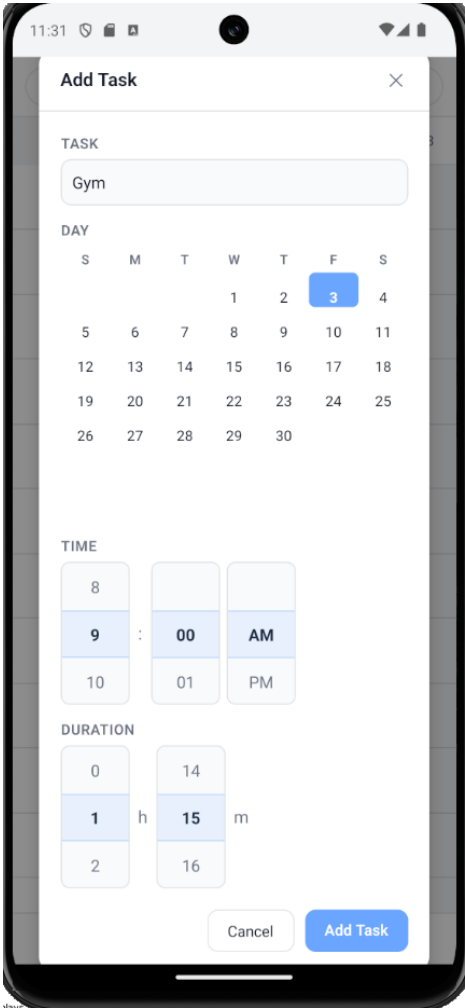


Fig.4 Add Task Screen

HELP & SUPPORT

FAQ

FAQ Item 1
FAQ Item 2
FAQ Item 3

Contact

Name

Email

Message

SUBMIT

Fig.5 Help & Support Screen

6.0 Software Requirements Matrix

SRS Req. ID	Paragraph Title	Component Number	Component Name
FUNC_SRS_001	Launch Screen	6.2.1	Login / Launch Screen Wireframe
FUNC_SRS_002	User Authentication	5.2.1	Authentication Component
FUNC_SRS_003	User Registration	5.2.1	Authentication Component
FUNC_SRS_004	Calendar Integration	5.2.4	Calendar Component
FUNC_SRS_004.1	Calendar will provide reminders and notifications	5.2.5	Reminder & Notification Component
FUNC_SRS_005	AI Scheduling and Task Planning	5.2.6	AI Scheduling Component
FUNC_SRS_005.1	Users can input task name, day, start time, duration hours, and duration minutes	5.2.2, 5.2.3	User Profile & Preferences, Task Management
FUNC_SRS_005.2	Task draft generation and task creation	5.2.6	AI Scheduling Component
FUNC_SRS_006	Help and Support Section	5.2.7	Help & Support Component
FUNC_SRS_006.1	Help section (FAQs, tutorials, troubleshooting)	6.2.5	Help & Support Screen
FUNC_SRS_006.2	Contact Us feature (support requests and feedback)	6.2.5	Help & Support Screen
FUNC_SRS_007	Task Priority	5.2.3	Task Management Component
FUNC_SRS_008	AI conversation session	5.2.6	AI scheduling component
FUNC_SRS_009	Calendar state management	5.2.4	Calendar Component